

COMPLIANCE CHARACTERIZATION AND CONTROLLED EXPANSION EMBODIMENTS

In various embodiments, the device includes a compliance characterization region configured to quantify intraluminal tissue response, deformation behavior, expansion tolerance, or interaction pressure under controlled measurement conditions. Compliance characterization may be performed using non-inflatable sensing structures, inflatable expansion structures, or hybrid configurations combining multiple modalities.

In non-inflatable embodiments, one or more pressure sensing elements are positioned on or within the insertable measurement section to detect contact pressure against surrounding tissue. Pressure sensors may be distributed circumferentially, arranged longitudinally, or provided as segmented rings to produce spatially resolved pressure response profiles. Suitable sensing modalities may include force sensitive resistors (FSR), piezoresistive elastomers, capacitive pressure sensors, strain-based pressure structures, microfluidic pressure transmission systems coupled to MEMS pressure sensors, or combinations thereof.

In inflatable embodiments, the compliance characterization region includes at least one expandable cuff or inflatable structure configured to expand radially outward from the insertable section. Inflation may be performed by a manual pump, an automated pump, or a regulated inflation architecture, and may be applied to a localized ring region, multiple separated regions, or an extended region covering a substantial portion of the insertable section.

Inflation behavior may be monitored using one or more sensing methods including direct cuff pressure measurement via a pressure sensor, cuff expansion measurement via a strain sensor or displacement sensor, flow monitoring, valve state monitoring, actuator telemetry, or inferred expansion derived from pump input and pressure response dynamics. In certain embodiments, the device may control inflation to one or more target setpoints and record pressure-versus-time and/or expansion-versus-time characteristics to generate compliance curves.

Hybrid embodiments combine inflatable expansion with pressure sensing to simultaneously capture controlled expansion response and contact pressure distribution. In such embodiments, pressure sensors may be located beneath, within, or adjacent to the inflatable structure to detect contact behavior as the cuff expands.

In certain embodiments, the compliance characterization region may include relief structures or pressure limiting mechanisms configured to prevent over-inflation. Such safety mechanisms may include mechanical relief valves, compliant expansion limiters, firmware-based shutoff thresholds, or combinations thereof.

Compliance metrics may include one or more derived values such as expansion threshold, peak pressure at target expansion, stiffness indices, hysteresis characteristics, response time constants, stability metrics, symmetry measures, or repeatability scores. In certain embodiments, compliance metrics may be generated as abstracted indices configured for privacy-preserving display while maintaining quantitative meaning.

Data acquired by the compliance characterization region may be processed locally or transmitted to an external computational system for filtering, artifact rejection, spoof-resistance validation, and generation of user profile parameters or derived indices suitable for integration into broader physiologic analytics platforms.